PAGE 1: KHARIF SEASON AGRONOMY & CROPPING CALENDAR
Definition & Overview: Kharif season, also known as the monsoon cropping season, begins wi
Key Kharif Crops: Rice \(Paddy\), Maize \(Corn\), Cotton, Soybean, Groundnut, Pigeonpe
Sowing Windows & Monsoon Dynamics:
1. North India \(Punjab, Haryana, UP\): Sowing window is June 15 to July 15. For transpl
2. Central & Western India \(MP, Maharashtra, Gujarat\): Sowing commences with the first
3. Eastern & Southern India: Sowing spans June 1 to July 31 depending on monsoon advanceme
Critical Agronomic Management:
- Land Preparation: Deep summer ploughing followed by 2 cross-harrowings to destroy weed s
- Drainage Planning: Provide surface drainage channels \(slope 0.1-0.2%\) to prevent wat
- Seed Priming & Fungicide Inoculation: Treat all kharif seeds with Trichoderma viride @ 5

PAGE 2: RABI SEASON AGRONOMY & WINTER CROPPING

Definition & Dynamics: Rabi cropping season spans winter and spring, sown from October to

Major Rabi Crops: Wheat, Rapeseed & Mustard, Chickpea \(Bengal Gram\), Field Pea, Lentil

Temperature Requirements & Vernalization:

- Germination Temperature: 20℃ - 25℃.
- Tillering & Vegetative Stage: Cool weather with temperatures between 10℃ - 15℃ promoti
- Grain Filling & Ripening: Warm, clear sunny days \(20℃ - 28℃\). Avoid terminal heat

Agronomic Interventions:

- Timely Sowing: Sowing wheat between November 1 and November 20 maximizes yield potential
- Crown Root Initiation \(CRI\) Irrigation: First irrigation must be applied precisely a
- Pre-sowing Seed Treatment: Treat chickpea and pulse seeds with Rhizobium culture and Pho

PAGE 3: ZAID \(SUMMER\) SEASON CROPPING & HEAT ADAPTATION
Definition & Opportunity: Zaid season is the short summer cropping window between Rabi har
Major Zaid Crops: Summer Moong \(Green Gram - SML 668, IPM 205-7, Samrat\), Summer Urad
Agronomic Management Protocols:
1. Rapid Field Turnaround: Sow summer moong immediately after wheat or potato harvest by M

PAGE 4: AGRO-CLIMATIC ZONES OF INDIA & REGIONAL PLANNING

The Planning Commission delineated 15 distinct Agro-Climatic Zones based on physiography,

1. Western Himalayan Region \(J&K, HP, Uttarakhand\): Apple, saffron, walnut, temperate
2. Eastern Himalayan Region \(Assam, NE States, Sub-Himalayan WB\): Tea, rice, pineapple
3. Lower Gangetic Plains \(West Bengal\): Rice-rice and rice-jute cropping systems, fish
4. Middle Gangetic Plains \(UP, Bihar\): Rice-wheat, maize, sugarcane, pigeonpea, mustar
5. Upper Gangetic Plains \(Western UP\): High-intensity rice-wheat, sugarcane, potato, m
6. Trans-Gangetic Plains \(Punjab, Haryana, Delhi, Rajasthan Ganganagar\): Highly mechan
7. Eastern Plateau & Hills \(Chota Nagpur, Odisha, MP\): Rainfed rice, minor millets, ni
8. Central Plateau & Hills \(Bundelkhand, Malwa, MP\): Soybean, wheat, chickpea, mustard
9. Western Plateau & Hills \(Maharashtra\): Cotton, sugarcane, jowar, bajra, pomegranate
10. Southern Plateau & Hills \(AP, Telangana, Karnataka, TN\): Groundnut, cotton, ragi,
11. East Coast Plains \(Coastal AP, TN, Odisha\): Coastal paddy, coconut, cashew, oil pa
12. West Coast Plains & Ghats \(Kerala, Coastal Karnataka, Goa\): Spices, rubber, coconu
13. Gujarat Plains & Hills: Cotton, groundnut, castor, cumin, sesame, bajra.
14. Western Dry Region \(Western Rajasthan\): Pearl millet, cluster bean \(guar\), mot
15. Island Region \(Andaman & Nicobar, Lakshadweep\): Coconut, arecanut, tropical fruits

PAGE 5: SCIENTIFIC CROP SELECTION FRAMEWORK
The 6-Pillar Decision Framework for Crop & Variety Selection:
Pillar 1: Soil Characteristics:
- Heavy Clay Soils \(Vertisols\): High water retention; ideal for Cotton, Soybean, Wheat
- Sandy Loam Soils \(Inceptisols/Entisols\): Excellent aeration; ideal for Potato, Groun
- Saline/Alkaline Soils: Choose tolerant crops \(Barley, Mustard, Cotton, Beetroot\).
Pillar 2: Water Availability & Irrigation Source:

PAGE 6: CLIMATE CHANGE ADAPTATION & NICRA TECHNOLOGIES
National Innovations in Climate Resilient Agriculture \(NICRA\) Strategic Interventions:
1. Flash-Flood & Submergence Adaptation:
- Deploy Swarna-Sub1, Samba Mahsuri-Sub1, and CR Dhan 801 rice varieties capable of surviv
2. Drought Adaptation & Moisture Scarcity:
- Use Sahbhagi Dhan, CR Dhan 201, and DRR Dhan 42 for direct dry seeding, maturing in 110-
3. Terminal Heat Stress in Wheat:

PAGE 7: CROP DIVERSIFICATION & INTERCROPPING RATIOS
Intercropping maximizes Land Equivalent Ratio \(LER > 1.25\), reduces economic risk, and
Standard Recommended Intercropping Combinations & Spatial Geometries:
1. Wheat + Mustard \(8:1 or 9:1 row ratio\):
- 8 rows of Wheat \(PBW 824 / HD 3086\) spaced at 20 cm alternating with 1 row of Mustar
2. Chickpea + Mustard \(6:1 or 4:1 row ratio\):
- Chickpea \(JG 14\) fixes atmospheric nitrogen, reducing mustard fertilizer needs.
3. Maize + Cowpea / Soybean \(1:1 or 2:2 row ratio\):
- Provides balanced cereal-legume fodder and grain while smothering weeds.
4. Sugarcane + Potato / Onion / Garlic \(1:2 row ratio\):
- Short-duration crops grown in inter-row space \(90-120 cm\) of autumn sugarcane provid
5. Cotton + Green Gram / Black Gram \(1:2 row ratio\):
- Legumes cover the ground during initial 60 days of slow cotton growth, reducing weed eme

PAGE 8: SUSTAINABLE CROP ROTATION CYCLES

Monoculture \(continuous Rice-Wheat or Cotton-Cotton\) leads to groundwater depletion, m

Recommended Sustainable Rotational Cycles:

Cycle 1: Rice - Wheat - Summer Moong \(3 crops/year\):

- Green gram incorporated into soil after 2 pod pickings adds 35-40 kg N/ha and breaks the

Cycle 2: Cotton - Wheat - Fallow \(2-year cycle\):

- Replaced with Cotton - Chickpea or Cotton - Mustard to conserve 40% winter irrigation.

Cycle 3: Maize - Mustard - Summer Moong \(Intensive Low-Water Rotation\):

- Consumes 60% less water than Rice-Wheat while providing equivalent net returns.

Cycle 4: Soybean - Wheat - Green Manuring \(Dhaincha\):

- Central India rotation optimizing soil organic carbon and reducing urea requirement by 2

Rotational Benefits:

- Reduces nematode populations by 70% when non-host crops \(Mustard, Marigold\) are rota
- Destroys Phalaris minor seed bank in wheat when replaced with Berseem or Sunflower.

PAGE 9: NATURAL FARMING & BIO-INPUT PREPARATION

Zero Budget Natural Farming \(ZBNF\) is based on 4 Core Pillars:

1. Jeevamrutha \(Liquid Microbial Bio-Enhancer\):

- Recipe: 200 L water + 10 kg fresh indigenous desi cow dung + 5-10 L desi cow urine + 2 k
- Fermentation: Stir clockwise twice daily in shade for 48-72 hours.
- Application: Apply 200 L/acre through irrigation water or as 10% foliar spray every 15-2

2. Beejamrutha \(Seed Microbial Shield\):

- Recipe: 20 L water + 5 kg desi cow dung + 5 L cow urine + 50 g slaked lime \(chuna\) +
- Application: Slurry coat on seeds before sowing; prevents seed-borne and soil-borne fung

3. Achhadana \(Mulching\):

- Soil mulching with dry crop residue \(3-4 tonnes/acre\) keeps root zone cool, conserve

4. Whapasa \(Soil Moisture-Aeration Equilibrium\):

- Water is applied only to alternate furrows during afternoon, creating vapor aeration rat

PAGE 10: AGROFORESTRY & BOUNDARY PLANTATION
Agroforestry integrates woody perennials with agricultural crops on the same land manageme
Agroforestry Models:
1. Poplar \(Populus deltoides\) + Agri-Crops \(North India\):
- Spacing: 5 m x 4 m or 7 m x 3 m \(500 trees/ha\).
- Compatible Crops: Sugarcane, Wheat, Turmeric, Ginger, Potato during initial 3 years. Dec
- Economics: 5-year timber harvesting generates Rs 4,00,000 - 6,00,000/acre net timber rev
2. Melia dubia \(Malabar Neem\) + Pulses/Vegetables \(South & Central India\):
- Fast-growing plywood tree harvesting in 6-7 years. Tolerates high temperature and provid
3. Windbreak & Shelterbelt Design:
- Plant 2-3 staggered rows of Casuarina, Bamboo, or Shisham along the south-western bounda
- Reduces wind velocity by 50-60%, slashing crop evapotranspiration loss by 20-25% and pre

PAGE 11: SOIL TESTING METHODOLOGY & SOIL HEALTH CARD

Soil testing is the scientific assessment of soil nutrient availability, chemical balance,

Representative Soil Sampling Protocol:

1. Sampling Pattern: Divide farm into uniform management units based on slope, color, and
2. Sampling Depth: 0-15 cm for field crops \(cereals, pulses, oilseeds\); 0-30 cm for de
3. Sample Processing: Mix sub-samples thoroughly, quarter down to 500 grams, shade dry, cr

The 12 Critical Soil Health Card Parameters:

PAGE 12: SOIL pH & ACID SOIL RECLAMATION

Soil Reaction \(pH\) Dynamics: Soil pH measures hydrogen ion activity. Optimal nutrient

Acid Soil Constraints \(pH < 6.0\):

- Aluminum \(Al3+\) and Manganese \(Mn2+\) toxicity causes root stunting and ’stubby r
- Phosphorus gets chemically fixed as insoluble Aluminum and Iron Phosphates, resulting in
- Beneficial bacterial populations \(Rhizobium, Nitrosomonas\) drop by up to 80%.

Reclamation & Liming Protocols:

PAGE 13: SALINE & SODIC SOIL RECLAMATION
Classification of Salt-Affected Soils:

PAGE 14: SOIL ORGANIC CARBON \(SOC\) ENHANCEMENT
Soil Organic Carbon \(SOC\) is the fundamental indicator of soil biological vitality, ca
Proven SOC Enhancement Strategies:
1. Farmyard Manure \(FYM\) & Well-Decomposed Compost:
- Apply 8-10 tonnes/ha well-rotted FYM \(C:N ratio 20:1 to 25:1\) 3 weeks before sowing.
2. Vermicomposting \(Eisenia fetida Earthworms\):
- Apply 2.5 - 3.5 tonnes/ha vermicompost. Contains 5x available N, 7x available P, 11x ava
3. In-situ Green Manuring:
- Sow Dhaincha \(Sesbania aculeata\) or Sunnhemp \(Crotalaria juncea\) @ 50 kg/ha with
- Plough down at 45-50 days \(flowering stage\). Adds 20-25 tonnes/ha succulent green bi
4. Retention of Crop Residues \(Zero-Burn Policy\):
- Retaining wheat/paddy stubble \(5-7 tonnes/ha\) using Super Seeder or Happy Seeder pre

PAGE 15: SOIL COMPACTION & SUBSOILING

Soil Physical Matrix & Compaction Hazards:

- Ideal Soil Composition: 45% mineral matter, 5% organic matter, 25% soil water, 25% soil
- Subsoil Hardpan Formation: Continuous rotary tillage at 10-15 cm and repeated heavy trac
- Consequences: Root penetration stops, water infiltration drops by 75%, causing waterlogg

Diagnostic & Remediation Protocol:

1. Penetrometer Diagnosis: A cone penetrometer resistance exceeding 2.0 MPa \(300 PSI\)
2. Subsoiling \(Chisel Ploughing\):

- Equipment: Single or multi-shank subsoiler penetrating 45-60 cm depth.
- Spacing: Run subsoiler at 1.0 to 1.5 meter spacing across the field in dry summer condit
- Frequency: Once every 3 to 4 years.

3. Yield Impact: Shattering hardpan increases cotton root depth from 30 cm to 90 cm, boost

PAGE 16: LASER LAND LEVELLING & SEEDBED ENGINEERING
Laser Land Levelling \(LLL\) Technology:
- Working Principle: A tractor-mounted scraper guided by a rotary laser transmitter and re
Quantified Agronomic & Economic Benefits:
1. Water Savings: Reduces irrigation water requirement by 25-30% due to rapid, uniform wat
2. Cultivable Area Increase: Eliminates unnecessary internal bunds and irrigation channels
3. Fertilizer Efficiency: Enhances fertilizer use efficiency by 15-20% by preventing nutri
4. Yield Boost: Produces uniform germination, synchronous tillering, and an average yield
5. Service Life: A single laser levelling remains effective for 3-4 years under standard t

PAGE 17: CONSERVATION AGRICULTURE & DIRECT SEEDED RICE \(DSR\)

Conservation Agriculture \(CA\) is defined by 3 Core Principles: \(1\) Minimum mechani

Zero-Till Wheat Sowing:

- Direct sowing of wheat immediately after paddy combine harvest without any field prepara
- Saves Rs 2,500 - 3,500/ha in diesel tillage costs and allows 10-12 days earlier sowing,

Direct Seeded Rice \(DSR\) Technology \(Tar-Watter Method\):

1. Land Preparation: Laser level field, apply pre-sowing heavy irrigation \(Rauni\), and
2. Sowing: Drill primed seed @ 20-25 kg/ha at 3-4 cm depth with DSR machine equipped with
3. Weed Management \(Critical\):

- Spray Stomp 30% EC \(Pendimethalin\) @ 1.0 L/acre within 24 hours of sowing.
- Follow with Nominee Gold \(Bispyribac-sodium 10% SC\) @ 100 ml/acre or Almix @ 8 g/acr

4. Water Saving: Conserves 20-25% groundwater by eliminating continuous puddling and pondi

PAGE 18: SOIL MICROBIOLOGY & BIOFERTILIZER SCIENCE

Beneficial soil microorganisms convert unavailable soil mineral reserves into plant-absorb

1. Nitrogen-Fixing Biofertilizers:

- Rhizobium spp.: Symbiotic N-fixer for legumes \(Chickpea, Soybean, Moong, Groundnut\).

PAGE 19: SOIL MICRONUTRIENT DEFICIENCY & REMEDIATION

Intensive agriculture has led to widespread multi-micronutrient deficiencies across 45% of

Specific Micronutrient Diagnostics & Soil/Foliar Remedies:

1. Zinc \(Zn\) Deficiency \(Khaira Disease in Rice / White Bud in Maize\):
- Symptoms: Rusty brown pigmentation on middle leaves, stunted internodes, delayed maturit
- Soil Application: Apply Zinc Sulfate Heptahydrate \(21% Zn\) @ 25 kg/ha or Zinc Sulfat
- Foliar Emergency: Spray 0.5% Zinc Sulfate \(5 g/L\) + 0.25% Lime \(2.5 g/L\) or 0.1%

2. Iron \(Fe\) Chlorosis:
- Symptoms: Interveinal chlorosis on youngest leaves turning completely bleached white. Co
- Foliar Remedy: Spray 1.0% Ferrous Sulfate \(FeSO4 19% Fe @ 10 g/L\) + 0.1% Citric Acid

3. Boron \(B\) Deficiency \(Hollow Heart in Groundnut / Cracking in Fruit\):
- Soil: Apply Borax \(10.5% B\) @ 10 kg/ha or Solubor \(20% B\) @ 5 kg/ha basal.
- Foliar: Spray 0.1-0.15% Solubor \(1.0-1.5 g/L\) at flower pre-bloom.

4. Sulfur \(S\) Deficiency \(Yellowing of Young Leaves in Mustard/Pulses\):
- Soil: Apply Agricultural Gypsum @ 250 kg/ha or Bentonite Sulfur 90% @ 25 kg/ha at sowing

PAGE 20: SOIL & WATER CONSERVATION ENGINEERING

Soil erosion strips topsoil rich in organic matter and nutrients, causing land degradation

Engineering & Agronomic Soil Conservation Structures:

1. Contour Bunding & Graded Bunding:

- Suitable for slopes between 2% and 6%. Earthen embankments constructed along elevation c
- Impounds runoff, allowing 80% rainwater percolation and reducing soil loss from 25 t/ha/

2. Bench Terracing:

- Essential on steep slopes \(>15% in hilly states\). Converts sloping land into a serie

3. Vegetative Barriers \(Vetiver Grass - Vetiveria zizanioides\):

- Planting dense hedge rows of Vetiver grass on contours at 1-meter vertical drops creates

4. Gully Plugging & Check Dams:

- Construct loose boulder check dams and gabion structures across drainage gullies to redu

PAGE 21: PRECISION AGRICULTURE & SMART FARMING PILLARS
Definition & Core Principles: Precision Agriculture \(Smart Farming\) is a data-driven f
Key Pillars & Technologies:
1. Global Navigation Satellite Systems \(GNSS\) & RTK Auto-Steer: Centimeter-level posit
2. Variable Rate Technology \(VRT\): Electro-hydraulic control of seed and fertilizer ap
3. IoT Sensor Telemetry: Soil moisture, EC, and leaf temperature sensors transmitting fiel
4. Drone Multispectral Imaging: High-resolution NDVI monitoring for stress detection 7-10
5. Edge AI Agronomy Decision Engines: Predictive algorithms calculating daily crop water b

PAGE 22: AUTOMATED FERTIGATION & WATER-SOLUBLE FERTILIZERS

Definition & Overview: Fertigation is the precise injection of 100% water-soluble fertiliz

Compatible Water-Soluble Formulations:

- 19:19:19 \(Starter/Vegetative\)
- 12:61:0 \(Mono Ammonium Phosphate MAP - Root initiation & flowering\)
- 13:0:45 \(Potassium Nitrate - Fruit enlargement & grain filling\)
- 0:52:34 \(Mono Potassium Phosphate MKP - Reproductive maturity & disease resistance\)
- 0:0:50+17.5%S \(Sulfate of Potash SOP - Quality, brix sugar, and color\).

Fertigation Compatibility Rules:

PAGE 24: ALTERNATE WETTING AND DRYING \(AWD\) WATER PROTOCOL
AWD Protocol Details: Developed by IRRI, AWD is an efficient irrigation strategy where fie
Implementation Steps:
1. Install a ’Pani Pipe’ \(30 cm long, 10-15 cm diameter perforated plastic pipe\) in th
2. After transplanting, keep 2-5 cm water for 2 weeks to establish roots.
3. Thereafter, allow water to subside. When water level in pipe drops to 15 cm below soil

PAGE 25: CRITICAL CROP GROWTH STAGES FOR IRRIGATION

Moisture stress at critical phenological stages causes irreversible yield loss:

1. Wheat: \(a\) Crown Root Initiation CRI \(21 DAS - most critical\), \(b\) Tillerin
2. Rice: \(a\) Seedling establishment, \(b\) Active Tillering, \(c\) Panicle Initiat
3. Maize: \(a\) Tasseling \(VT\) and \(b\) Silking \(R1\). Stress at silking cause
4. Cotton: \(a\) Squaring, \(b\) Peak Flowering, \(c\) Early Boll Development.
5. Chickpea: \(a\) Pre-flowering branching, \(b\) Pod development. DO NOT irrigate at

PAGE 26: SUB-SURFACE DRIP IRRIGATION \(SDI\)
SDI Architecture: Drip laterals embedded 15 to 30 cm below soil surface directly in the cr
Key Advantages:
- Zero evaporation loss from soil surface.
- Zero weed germination between crop rows as topsoil remains dry.
- Allows uninterrupted tractor and equipment movement on dry surface.
- Reduces lateral degradation from UV radiation and rodent damage.
- Delivers 95% water and nutrient application efficiency.
Technical Requirements: Heavy-duty pressure compensating \(PC\) emitters with anti-sipho

PAGE 27: IoT SOIL MOISTURE SENSING & AUTOMATION
Sensor Technologies:
1. Frequency Domain Reflectometry \(FDR\) & Capacitance Probes: Measure dielectric permi
2. Granular Matrix Sensors \(Watermark\): Measure soil water tension in centibars/kPa.
Irrigation Trigger Thresholds:

PAGE 28: PROTECTED CULTIVATION & HYDROPONIC SYSTEMS
Protected Cultivation Structures:
1. Naturally Ventilated Polyhouse \(NVPH\): G.I. pipe frame covered with 200-micron UV-s
2. Climate-Controlled Greenhouse: Evaporative cooling pad-and-fan system with foggers, kee
Hydroponic Soilless Systems:
- Nutrient Film Technique \(NFT\): Thin film of recirculating nutrient solution over pla
- Dutch Bucket System: Bato buckets filled with perlite/cocopeat substrate with automated
- Water Savings: Uses 90% less water than traditional open-field farming.

PAGE 29: SOLAR IRRIGATION & PM-KUSUM SCHEME
Solar Pumping Architecture: Photovoltaic \(PV\) array \(3 HP to 10 HP\) connected to a
PM-KUSUM Government Scheme Components:
- Component A: Decentralized solar power plants \(0.5 to 2 MW\) on barren farmlands.

PAGE 30: POST-HARVEST STORAGE & COLD CHAIN
Post-Harvest Loss Mitigation Technologies:
1. Grain Hermetic Storage \(PICS Bags / Cocoons\): Multi-layer polyethylene hermetic bag
2. IoT-Monitored Grain Silos: Temperature and humidity sensor cables inside galvanized cor
3. Solar-Powered Micro Cold Rooms: 5-10 MT cold rooms operating on thermal energy storage

PAGE 31: BENEFITS OF MODERN FARMING & AGRITECH ADOPTION

Key Benefits of Modern Farming Technologies:

1. Resource Efficiency: Drip irrigation saves 40-60% water while automated fertigation red
2. Yield Optimization: Precision farming tools and hybrid seed selection yield 20-35% high
3. Labor & Time Savings: Drone UAV spraying completes 1 acre in 7-10 minutes compared to 3

PAGE 32: SOIL HEALTH CARD PARAMETERS & AMELIORATION
Soil Health Card 12 Key Indicators & Chemistry Benchmarks:
1. pH \(1:2.5 soil-water\): Strongly Acidic \(<5.5\), Slightly Acidic \(5.5-6.5\), N

PAGE 33: PLANT NUTRIENT DYNAMICS & ADVANCED NUTRITION
Core Nutrient Principles: Plant nutrition requires balanced availability of 17 essential e
Fertilizer Dosage & Placement:
- Basal Application: Apply 100% of Phosphorus and Potassium alongside 25-33% Nitrogen at s
- Top Dressing: Apply remaining Nitrogen in 2-3 equal splits at active tillering/branching
- Foliar Supplementation: Spray 1% 19:19:19 or 0.5% chelated zinc during vegetative growth
- Soil Health Integration: Combine inorganic fertilizers with 5-10 tonnes/ha FYM to enhanc

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PAGE 41: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Seed Germination & Emergence
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 42: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Active Vegetative & Tillering
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 43: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Stem Elongation & Jointing
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 44: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Panicle Initiation & Booting
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 45: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Flowering & Anthesis
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 46: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Fruit Set & Pod Development
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 47: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Grain Filling & Milk Stage
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 48: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Dough & Maturity Stage
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 49: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Harvesting Indices & Threshing
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 50: CROP GROWTH STAGES & AGRONOMIC INTERVENTIONS
Phenological Milestone: Post-Harvest Curing & Moisture
Agronomic Characterization:
- Physiological Activity: Plants undergo rapid cellular division, carbohydrate allocation,
- Critical Resource Needs: Ensure adequate moisture without waterlogging. Moisture stress

PAGE 51: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
Weed Control Protocols:
1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowi
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland r
3. Chemical Control:
- Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
- Post-Emergence Herbicides: Apply Bispyribac-sodium 10% SC @ 100 ml/acre in rice or Clodi

PAGE 52: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
Weed Control Protocols:
1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowi
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland r
3. Chemical Control:
- Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
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PAGE 53: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
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PAGE 54: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
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PAGE 55: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
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3. Chemical Control:
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PAGE 56: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
Weed Control Protocols:
1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowi
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland r
3. Chemical Control:
- Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
- Post-Emergence Herbicides: Apply Bispyribac-sodium 10% SC @ 100 ml/acre in rice or Clodi

PAGE 57: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
Weed Control Protocols:
1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowi
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland r
3. Chemical Control:
- Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
- Post-Emergence Herbicides: Apply Bispyribac-sodium 10% SC @ 100 ml/acre in rice or Clodi

PAGE 58: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
Weed Control Protocols:
1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowi
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland r
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- Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
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PAGE 59: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
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2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland r
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- Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
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PAGE 60: INTEGRATED WEED MANAGEMENT \(IWM\) & HERBICIDE ROTATION
Weed Control Protocols:
1. Cultural Control: Stale seedbed technique, smother intercropping, and high-density sowi
2. Mechanical Control: Cono-weeding in wet rice at 15 and 30 DAT; wheel hoeing in upland r
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- Pre-Emergence Herbicides: Apply Pendimethalin 30% EC @ 1.0 L/acre or Pretilachlor 50% EC
- Post-Emergence Herbicides: Apply Bispyribac-sodium 10% SC @ 100 ml/acre in rice or Clodi

PAGE 61: COMPREHENSIVE PRODUCTION PACKAGE FOR RICE \(PADDY\)

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 62: COMPREHENSIVE PRODUCTION PACKAGE FOR WHEAT

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 63: COMPREHENSIVE PRODUCTION PACKAGE FOR COTTON

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 64: COMPREHENSIVE PRODUCTION PACKAGE FOR TOMATO

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 65: COMPREHENSIVE PRODUCTION PACKAGE FOR MAIZE \(CORN\)

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 66: COMPREHENSIVE PRODUCTION PACKAGE FOR POTATO

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 67: COMPREHENSIVE PRODUCTION PACKAGE FOR MUSTARD \(RAPESEED\)
1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 68: COMPREHENSIVE PRODUCTION PACKAGE FOR CHILLI \(CAPSICUM\)

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 69: COMPREHENSIVE PRODUCTION PACKAGE FOR SUGARCANE

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 70: COMPREHENSIVE PRODUCTION PACKAGE FOR SOYBEAN

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 71: COMPREHENSIVE PRODUCTION PACKAGE FOR CHICKPEA \(GRAM\)

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 72: COMPREHENSIVE PRODUCTION PACKAGE FOR GROUNDNUT

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 73: COMPREHENSIVE PRODUCTION PACKAGE FOR ONION & GARLIC

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 74: COMPREHENSIVE PRODUCTION PACKAGE FOR MILLETS \(BAJRA/RAGI\)
1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 75: COMPREHENSIVE PRODUCTION PACKAGE FOR EXOTIC VEGETABLES

1. Climate & Soil: Well-drained fertile loam with pH 6.0 to 7.8.
2. High-Yielding Varieties: Certified seeds with >98% purity and >85% germination rate.
3. Sowing / Planting: Follow recommended spacing and seed rate. Treat seed with fungicide
4. Fertilizer Schedule \(NPK kg/ha\): Apply balanced N:P2O5:K2O according to soil test c
5. Water Management: Irrigate at critical growth milestones, avoiding waterlogging and ext
6. Plant Protection: Follow Integrated Pest Management \(IPM\) with Economic Threshold L
7. Expected Yield: High yield with good management practices.

PAGE 76: INTEGRATED PEST & DISEASE MANAGEMENT - IPM PRINCIPLES & ECONOMIC THRE
Management Framework:
- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 77: INTEGRATED PEST & DISEASE MANAGEMENT - BIOLOGICAL PARASITOIDS & PREDA
Management Framework:
- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 78: INTEGRATED PEST & DISEASE MANAGEMENT - PHEROMONE & LIGHT TRAPPING SY
Management Framework:
- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 79: INTEGRATED PEST & DISEASE MANAGEMENT - SUCKING PEST COMPLEX MANAGEM

Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 80: INTEGRATED PEST & DISEASE MANAGEMENT - LEPIDOPTERAN BORER MANAGEME

Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 81: INTEGRATED PEST & DISEASE MANAGEMENT - MAJOR FUNGAL PATHOGEN CONTR
Management Framework:
- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 82: INTEGRATED PEST & DISEASE MANAGEMENT - BACTERIAL DISEASES & BACTERICI
Management Framework:
- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 83: INTEGRATED PEST & DISEASE MANAGEMENT - VIRAL VECTORS & BARRIER MANAG
Management Framework:
- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 84: INTEGRATED PEST & DISEASE MANAGEMENT - ROOT-KNOT NEMATODE AMELIORA
Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 85: INTEGRATED PEST & DISEASE MANAGEMENT - FUNGICIDE RESISTANCE MANAGEM

Management Framework:

- Monitoring & ETL Thresholds: Regularly scout fields; intervene with certified chemicals
- Biological Controls: Release Trichogramma egg parasitoids @ 50,000/ha or spray Beauveria
- Chemical Interventions: Use selective insecticides/fungicides at recommended label doses
- Spray Quality: Use 150-200 liters water/acre with hollow cone nozzles for insecticides a

PAGE 86: FARM SAFETY & PRECAUTIONS - FARMING SAFETY & PRECAUTIONS OVERVIEW

Essential Farming Precautions & Safety Guidelines:

1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(PPE\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 87: FARM SAFETY & PRECAUTIONS - PERSONAL PROTECTIVE EQUIPMENT \(PPE\) FOR
Essential Farming Precautions & Safety Guidelines:
1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(PPE\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 88: FARM SAFETY & PRECAUTIONS - RE-ENTRY INTERVAL \(REI\) & PRE-HARVEST INTE

Essential Farming Precautions & Safety Guidelines:

1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(PPE\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 89: FARM SAFETY & PRECAUTIONS - CHEMICAL STORAGE & SAFE DISPOSAL PROTOCO

Essential Farming Precautions & Safety Guidelines:

1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(PPE\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 90: FARM SAFETY & PRECAUTIONS - PESTICIDE TOXICITY CLASSIFICATIONS & FIRST A

Essential Farming Precautions & Safety Guidelines:

1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(PPE\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 91: FARM SAFETY & PRECAUTIONS - FARM MACHINERY & TRACTOR SAFETY PROTOCO
Essential Farming Precautions & Safety Guidelines:
1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(PPE\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 92: FARM SAFETY & PRECAUTIONS - EXTREME WEATHER PRECAUTIONS & LABOR SAF
Essential Farming Precautions & Safety Guidelines:
1. Chemical Spraying Precautions: Always wear Personal Protective Equipment \(PPE\) incl
2. Storage & Labeling: Store all pesticides in original labeled containers inside locked,

PAGE 93: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - PRECISION AGRICULTURE & Y
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

PAGE 94: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - AGRICULTURAL DRONES \(UA
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

PAGE 95: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - IOT WIRELESS SOIL & WEATH
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

PAGE 96: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - SATELLITE REMOTE SENSING
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

PAGE 97: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - PROTECTED POLYHOUSE CU
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

PAGE 98: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - SMART COLD CHAIN & HERME
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

PAGE 99: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - PM-KUSUM SOLAR SCHEME &
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input

PAGE 100: MODERN AGRICULTURE TECHNOLOGY & SCHEMES - GOVERNMENT AGRICULTUR
Technological Implementation & Government Support:
- Innovation Description: Modern digital agriculture optimizes resource use through automa
- Operational Impact: Boosts crop yields by 20-35%, saves 40-60% water, and reduces input